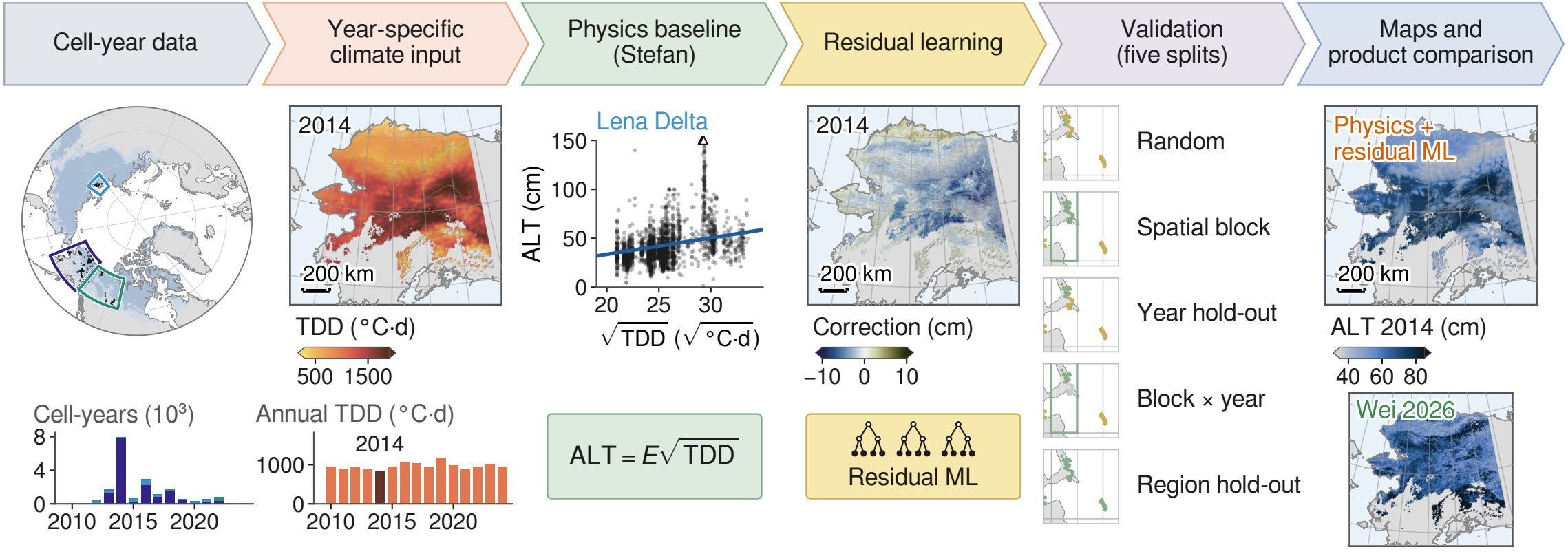

Cell-year data
Year-specific climate input
Physics baseline (Stefan)
Residual learning
Validation (five splits)
Maps and product comparison
2014
200 km
TDD (°C·d)
500 1500
Cell-years (10^3)
8
4
0
2010 2015 2020
Annual TDD (°C·d)
2014
1000
0
2010 2015 2020
Lena Delta
ALT (cm)
150
100
50
0
20 25 30
$\sqrt{\text{TDD}}$ ($\sqrt{°C·d}$)
$\text{ALT} = E\sqrt{\text{TDD}}$
2014
200 km
Correction (cm)
−10 0 10
Residual ML
Random
Spatial block
Year hold-out
Block × year
Region hold-out
Physics + residual ML
200 km
ALT 2014 (cm)
40 60 80
Wei 2026
Alaska · Lena Delta · Canada (NW)
Train
Test
Excluded